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Self-tuning Deep Brain Stimulation
External Globus Pallidus Observer
for Suppression of Beta Oscillations

by

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the degree of MSc Electronic Engineering

Abstract

Pathological beta oscillations are one of the major manifestations of Parkinson's disease (PD), and inhibiting their production is critical to the treatment of Parkinson's disease. Previous research has found that changes in the frequency of neuronal firing in the subthalamic nucleus (STN) and globus pallidus (GPe) regions lead to abnormal beta oscillations. The effects of the frequencies generated by the STN and GPe feedback loops on pathological beta oscillations have been explored deeply, but the specifics regarding neuronal activity in the GPe region are not well understood. Studying the state of neuronal activity in the GPe region by designing a suitable observer is beneficial for the further development of DBS for Parkinson's disease and is a starting point for closed-loop adaptive control for closed-loop treatment. The most notable innovation of this report lies in overcoming the traditional methods of dealing with delays in observers. A new type of observer was designed, which cleverly avoids the issues caused by delay estimation in the system, resulting in excellent observation outcomes.

This report constructs and compares, via simulation, several approaches to observer design (state-estimation) for a STN/GPe firing rate model. Observer construction is challenging since the firing rate model contains time delays and is inherently nonlinear. The approaches compared were the standard (linear) Luenberger observer, a recently introduced nonlinear observer, and a newly proposed nonlinear observer for time-delay systems. A key contribution of this MSc thesis is that it has established the theory of this observer, and this was previously unavailable in the literature. One of the challenges in observing GPe activity is the error and instability caused by delays. For the Luenberger observer and the nonlinear observer, delay is estimated using methods such as ignoring the delay, Padé approximation, and least squares approximation. One advantage of the advanced nonlinear observer (new nonlinear observer introduced here) is that it does not require delay approximation.

All observers were evaluated in simulation using Matlab and results compared using several measures, including the accuracy of the stable state, the correlation coefficient and the peak-to-peak ratio of the transitional state, and the RMSE after adding the AWGN (additive white gaussian noise). Overall, the evaluations revealed that traditional Luenberger designs were inadequate for state-estimates of the firing rate model. The recently introduced nonlinear observer was a little better, but the new observer proposed in this thesis was, by some margin, the most accurate and of sufficient quality to be used in the adaptive system.

Key words: DBS, Luenberger observer, nonlinear observer, advanced nonlinear observer, Padé approximation, least square approximation, RMSE, correlation coefficient

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Chapter 1 Introduction

1.1 Background

Parkinson's disease is a neurological disorder in which pathological beta oscillations are one of the main manifestations. Suppression of this abnormal oscillation is important in the treatment of Parkinson's disease. [1] It has been found that changes in the firing frequency of neurons in the subthalamic nucleus and the pallidum region led to the generation of beta oscillations. [2] And deep brain stimulation can be used to control these regions. Closed-loop brain stimulation offers advantages over traditional open-loop brain stimulation, but the controller may not perform optimally if operating conditions change. [3] These changes may include daily changes in symptoms, disease progression, and changes in electrode-tissue interface properties. To address this problem, studies have proposed adaptive or self-tuning control mechanisms that can accommodate slow changes in system characteristics over days, months, or years. [1] Such control mechanisms have the potential to provide long-term effective treatments. The effects of the frequencies generated by the STN and GPe feedback loops on pathological beta oscillations have been explored deeply, but the specifics regarding neuronal activity in the GPe region are not well understood. [3] Studying the state of neuronal activity in the GPe region by designing a suitable observer is beneficial for the further development of DBS for Parkinson's disease.

One of the challenges in observing the state of the GPe under adaptive DBS regulation based on the Firing-Rate model is the oscillations caused by time delays resulting from the limited transmission speed of axons and synapses between the two populations. [3] Since adaptive DBS regulation under the Firing-Rate model is a highly nonlinear model, the impact of delays on the design of the observer is significant. On one hand, it is necessary to design an observer that can match the delays and stabilize the final state. On the other hand, the observer needs to have good performance on this basis.

Another challenge is the control system of the model is nonlinear. To address the nonlinear characteristics of the adaptive DBS model under Firing-Rate, observers using two main approaches. One approach is to linearize the nonlinear structure and use a traditional Luenberger observer for observation. The other approach is to use a nonlinear observer directly. The Luenberger observer has a simple structure, while the nonlinear observer is more complex. However, both types of observers require certain methods to estimate delays, which significantly reduces the performance of the observer. Therefore, this report ultimately derives an advanced nonlinear observer model with delay characteristics based on the nonlinear observer proposed by Xingzhe Fan and Murat Arcak, [4] and proves that, asymptotically, the estimated states will approach the actual state. This model does not require delay estimation, thereby greatly improving the performance of the observer.

It should be emphasised here that although there are signal processing methods that can estimate the unknown state relatively accurately, these predictions are based on certain a priori probabilities, whereas the observer is valuable in that it does not need to rely on any prior knowledge of the data, but rather observes the unknown state in real time, which is its unique advantage.

1.2 Lyapunov Function

In a linear system, the stability of the system can be determined by analysing the eigenvalues of the system matrix. A system is asymptotically stable if the real part of all eigenvalues are negative. But in nonlinear systems it is not possible to study the stability of the system in this way. This is because the behaviour of nonlinear systems is very complex, and their complexity is divided into three main areas: diverse dynamic behaviour, local and global problems of stability, and nonlinear effects. Diverse dynamic behaviours such as periodicity, chaos, and limit rings. These behaviours do not exist in linear systems and eigenvalue analysis cannot capture these complex dynamic properties. Stability analyses of linear systems are usually global, namely if the system is stable at some point, it is stable throughout the state space. However, the stability of a nonlinear system may be local, namely the system is stable near a certain equilibrium point, but may be unstable in regions far from that equilibrium point. Eigenvalue analysis cannot deal with this problem of local stability. As for nonlinear effects, in nonlinear systems, small perturbations may cause large responses, and eigenvalue analysis, which assumes that the system behaviour is linear, cannot deal with these nonlinear effects. In contrast, the Lyapunov method can handle the complexity of nonlinear systems by constructing a Lyapunov function. [19] Specifically, the Liapunov method does not depend on the linearisation of the system and can be directly applied to non-linear systems. And the local and global stability of the system can be analysed by choosing appropriate Lyapunov functions. From the energy point of view, the Liapunov function is similar to the ‘energy’ of the system, and the stability of the system can be judged by analysing the change of energy.

It should be noted that the derivation of both the nonlinear observer and the advanced nonlinear observer is based on the second theorem of Lyapunov stability, which is related to the stability of solutions to differential equations. Consider a function $V(x): R^n \rightarrow R$ such that:

$$V(x) \geq 0, \text{ the equality holds only at } (x = 0) \text{ (positive definite function)}. \quad (1a)$$

$$\dot{V}(x(t)) < 0 \text{ (negative definite)} \quad (1b)$$

then $V(x)$ is called a Lyapunov function candidate, [19] and the system (from Lyapunov’s perspective) is asymptotically stable. In the above equation, a positive definite function simply means that the function takes 0 at the origin and positive values at any other point. The opposite is true for a negative definite function, which means that the function takes 0 at the origin and negative values at any other point. If the system is stable, the energy of the system decreases with time and eventually converges to a stable value. The above equation can be simply compared to the concept in physics where energy decreases over time and the reduced energy does not recover. Therefore, the system will eventually come to rest at a specific state, achieving stability.

1.3 Scope

The goal of this project is to achieve optimal performance in observing the GPe state by designing suitable observers. Although the traditional Luenberger observer has a simple structure and fast convergence, it cannot effectively capture transition state information due to the highly nonlinear nature of the control system being observed. Nonlinear observers can capture transition state information to some extent, but they are potentially unstable due to inaccurate delay estimation and have very complex structures. Therefore, an advanced nonlinear observer is derived, which can capture transition state information well without being overly complex.

The specific focus of this project is:

1. To construct the STN/GPe model in Simulink
2. To design a Luenberger observer with/without delay approximations
3. To design a Fan and Arca observer with/without delay approximations
4. To develop theory for a new advanced nonlinear observer and apply it
5. To compare all approaches

This report is divided into five chapters based on the key research directions of this project. Chapter 1 introduces the background and overview of the project. Chapter 2 provides a literature review of self-tuning DBS models based on the Firing-Rate model and existing observer structures. [5] Chapter 3 discusses the specific structural design of the observers applied in this project and the handling of delays. Chapter 4 introduces the performance analysis of the observers. Chapter 5 discusses the simulation results and conclusions.

Current literature has extensively discussed the oscillatory relationship between STN and GPe, but there is almost no research on observing GPe. [3] The observer models constructed for GPe in this project may help future research on self-tuning DBS. The innovative advanced nonlinear observer proposed in this report also provides a new perspective for nonlinear observers in delayed control systems.

CHAPTER 2 Theory and Literature Review

2.1 Theory and background of Closed-loop DBS model

Deep brain stimulation (DBS) is a treatment method suitable for patients with advanced Parkinson's disease. Currently, the most commonly used clinical approach involves long-term implanted electrodes and subcutaneous pulse generators to stimulate the subthalamic nucleus and globus pallidus in an open-loop manner. [12] However, this method may lead to overstimulation and involves a time-consuming process of parameter retuning. Therefore, recent closed-loop DBS schemes have attracted the interest of many researchers. [13] In some closed-loop stimulation studies, the stimulation signal is delivered in an "on-off" manner, [14] turning the stimulation on or off based on whether biomarkers exceed a specified threshold. Subsequently, researchers proposed a proportional feedback method, [5] where the stimulation amplitude is proportional to the measured biomarkers, and clinical trials have confirmed its good therapeutic effects. However, like "on-off" stimulation, this method also requires readjustment when facing changes in system characteristics. To address this issue, researchers have proposed a self-tuning control strategy inspired by adaptive control theory, where the control gain evolves based on measured pathological activity. [5] This method has been mathematically proven through Lyapunov-Krasovskii analysis [15] in the firing rate model to ensure the disruption of pathological oscillations when the coupling within the GPe is sufficiently weak.

The STN-GPe loop model initially proposed by Nevado-Holgado et al. (2010) can be used to study the emergence of pathological β oscillations observed in the basal ganglia of Parkinson's disease. [5] The model is defined as follows:

$$\gamma_1 \dot{x}_1(t) = -x_1(t) + S_1(c_{11}x_1(t - \alpha_{11}) - c_{12}x_2(t - \alpha_{12}) + b_1u_1(t)) \quad (2a)$$

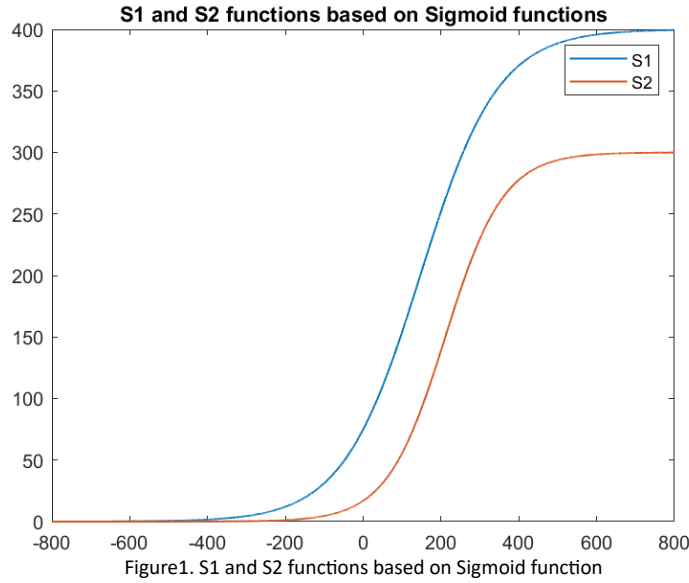
$$\gamma_2 \dot{x}_2(t) = -x_2(t) + S_2(c_{21}x_1(t - \alpha_{21}) - c_{22}x_2(t - \alpha_{22}) - b_2u_2(t)) \quad (2b)$$

In the above equation, S_1 and S_2 are the activation function, which simulates the response of a population of neurons to its input. The specific expression is:

$$S_i(x) = \frac{M_i B_i}{B_i + \exp\left(\frac{-4x_i}{M_i}\right)(M_i - B_i)} \quad (3)$$

where the constants were set to $M_1 = 300$, $M_2 = 400$, $B_1 = 17$, $B_2 = 75$. [5]

Specifically, as the following figure:



As shown in the above figure, this function is strictly increasing. Furthermore, if the starting point is restricted to be greater than 0, it is a strictly increasing function greater than 0. γ_1 and γ_2 are time constants, and for each $i, j \in \{1, 2\}$, the constants $c_{ij} \geq 0$ denote the strength of synaptic connections from population j to population i . $\alpha_{ij} \geq 0$ is the time delay that occurs due to the finite speed of axonal and synaptic transmission. $u_1(t)$ and $u_2(t)$ denote the effect of cortical (into the STN) and striatal (into the GPe) inputs to the system, respectively, as regulated by the synaptic weights $b_1 \geq 0$ and $b_2 \geq 0$.

Assuming that the cortical and striatal inputs are constants $((u_1(t), u_2(t))^T = \bar{u})$ consider the changes in variables $(u - \bar{u} \rightarrow u)$ and $(x - \bar{x} \rightarrow x)$, where $\bar{x}$ is the equilibrium value of $x = (x_1, x_2)^T$ for the input $\bar{u}$ whose existence is guaranteed by Pasillas-Lépine (2013). Move the activation function S_j so that $S_j(0) = 0$ for $j \in \{1, 2\}$, and add feedback $u(t)$ to represent the effect of artificial stimulation (DBS) on the STN, resulting in the following dynamic system:

$$\gamma_1 \dot{x}_1(t) = -x_1(t) + S_1(c_{11}x_1(t - \alpha_{11}) - c_{12}x_2(t - \alpha_{12}) + u(t)) \quad (4a)$$

$$\gamma_2 \dot{x}_2(t) = -x_2(t) + S_2(c_{21}x_1(t - \alpha_{21}) - c_{22}x_2(t - \alpha_{22})) \quad (4b)$$

$$u(t) = -\rho x_1(t) \quad (5)$$

where the proportional gain $\rho > 0$.

The proportional gain needs to be adjusted to an appropriate value so that the pathological oscillations decay to the expected steady state. Table 1 shows the corresponding parameters adopted in this experiment, [9,10,11,12], and the value of ρ is a reasonable value obtained through many experiments.

TABLE 1 PARAMETERS FOR CLOSED-LOOP FIRING-RATE DBS MODEL

γ_1	6ms
γ_2	14ms
c_{12}	10.7
c_{21}	20
c_{22}	12.3
α_{11}	0ms
α_{12}	7ms
α_{21}	1ms
α_{22}	1ms
$\bar{x}$	35

The data for the above parameters is derived from previous related studies, [9,10,11,12] referencing model parameter data for patients with completed Parkinson's disease. For patients with mild Parkinson's disease, the values are slightly different. If $\bar{x}$ and equation (4) are substituted into equation 3(a), [5] the following expression will be obtained:

$$6\dot{x}_1(t) = -x_1(t) + S_1(-25x_1(t) - 10.7x_2(t - 7) + 875) \quad (6)$$

That is, the value of ρ is not directly found here, but the coefficients are combined to find the appropriate parameter values after the combination, because the α_{11} is 0 here. It means that the sum of c_{11} and the ρ is always -25. Because in a real environment, there could be many different c_{11} values. The value 875 is considered after shifting the steady state $\bar{x}$ to the origin. The state figure of GPe under the Firing-Rate model with closed-loop proportional gain adjustment is as follows:

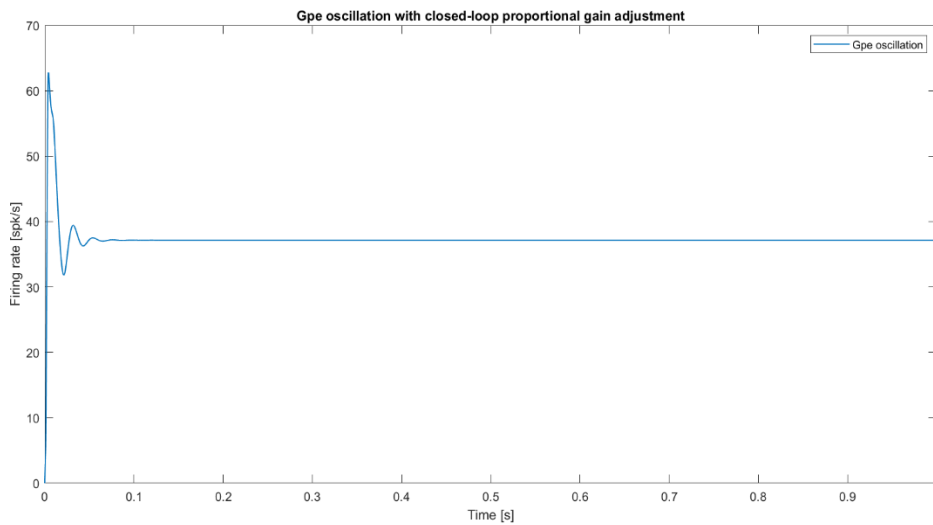


Figure2. Gpe oscillation with closed-loop proportional gain adjustment

From Figure 2, it can be seen that the intense oscillations gradually return to stability after the introduction of DBS, which is consistent with the oscillation situation of healthy people. And this behaviour is obtained with ‘perfect’ measurement of $x_2(t)$, whereas in reality we would like to estimate it.

2.2 Theory of Self-Tuning Closed-loop DBS model

As described in section 2.1, the proportional gain feedback method solves the drawbacks of open-loop and fixed stimulation feedback, but it is still weak in robustness to environmental changes, and each system change requires readjustment of the proportional parameters,[5] so this is not the best solution for closed-loop systems. On this basis, the self-regulating closed-loop system proposed by scholars is more reliable, that is, the feedback parameters can be automatically updated through the state of STN, and finally make the oscillations stable. [5] The specific formulas are as follows:

$$u(t) = -\rho(t)x_1(t) \quad (7a)$$

$$\gamma_\rho \dot{\rho}(t) = |x_1(t)| - \delta \rho(t) \quad (7b)$$

And after multiple experiments, the values of γ_ρ and δ are 0.3 and 0.7513 respectively. [5] By substituting the above two equations into equation (3a), and combining with equation 3(b), we can obtain the state figure of the adaptive control closed-loop system GPe.

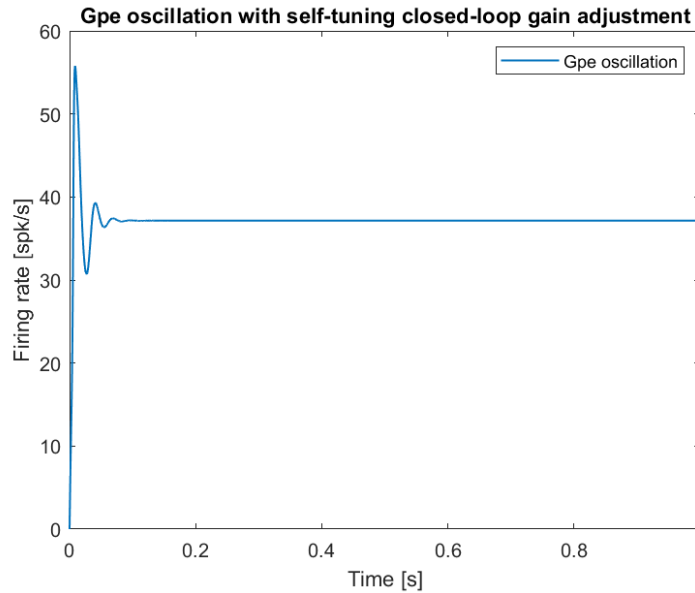


Figure3. Gpe oscillation with self-tuning closed-loop gain adjustment

From Figure 3, it can be seen that the pathological oscillation can still reach stability during the process of self-tuning feedback gain adjustment.

2.3 Theory of Observers

Observers can be classified into linear observers and nonlinear observers based on the simplicity of the structure of the controllers they need to observe. [16] The basic Luenberger observer is a linear observer. In continuous-time linear time-invariant systems, namely:

$$\dot{x} = Ax + Bu \quad (8a)$$

$$y = Cx \quad (8b)$$

The Luenberger observer has the following form:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}) \quad (9)$$

In the above equation, L is the gain matrix of the observer. And its observer error will meet the following equation:

$$\dot{e} = (A - LC)e \quad (10a)$$

$$e = \dot{x} - \dot{\hat{x}} \quad (10b)$$

The goal is to design an appropriate L such that by adjusting $A - LC$ to be a Hurwitz matrix, making all its eigenvalues are negative, so the observer's error can asymptotically converge to zero. However, it is obvious that the controller in this project is nonlinear. Therefore, if the Luenberger observer is to be used, the controller model needs to be approximated as a linear controller, which will obviously introduce a lot of errors. Hence, a more reliable observer is a nonlinear observer. [7] There are many forms of nonlinear observers, and this project first adopts the multivariable nonlinear observer proposed by Xingzhe Fan and Murat Arcak. [8] because their observer structure is almost compatible with our STN/GPe firing model. For the observer design, the plant is like:

$$\dot{x} = Ax + G\theta(Hx) + \mu(y, u) \quad (11a)$$

$$y = Cx \quad (11b)$$

The $\theta(\cdot)$ represents the nonlinear function. And with the above assumption, the observer has the form:[8]

$$\dot{\hat{x}} = A\hat{x} + L(C\hat{x} - y) + G\theta(H\hat{x} + K(C\hat{x} - y)) + \mu(y, u) \quad (12)$$

The target is to design the observer matrix L and K . And from the circle criterion and the properties of SPR, the matrix relationship that guarantees asymptotic stability can be derived.

That is:
$$\begin{bmatrix} (A + LC^T)P + P(A + LC) + \zeta I & PG + (H + KC)^T \\ G^TP + (H + KC) & 0 \end{bmatrix} \leq 0 \quad (13)$$

where $P = P^T > 0$, constant $\zeta > 0$. [8]

The equation (12) is a matrix inequality which can be solved to obtain the observer gain matrices L and K .

From functions 9(a) and 9(b), it is clear that this model is suitable for nonlinear systems without delay. However, the system in this project includes delays. Therefore, to use this observer, the delay needs to be estimated, and the state of the delay should be treated as a new state. The basic principles of delay estimation will be introduced in Section 2.4. Such a nonlinear observer will always be affected by delay errors. Consequently, this project has derived an observer based on this nonlinear observer that does not require delay estimation. This observer not only

has a simpler structure but also converges more easily. The structure of this observer will be detailed later in the text.

2.4 Theory of delay approximation

According to the basic theory of Laplace transform in signals and systems, a delayed signal $x(t - \alpha)$ can be represented in the frequency domain as: $x(s)e^{-\alpha s}$. Unfortunately, the Laplace transform cannot be directly applied to the model in this project because it cannot form integral or differential terms, and thus cannot construct the delay term into a new state equation. Therefore, the key to dealing with the delay is to approximate $e^{-\alpha s}$ as a structure of rational function, like $\frac{as+b}{cs+d}$. The reason for not considering the Taylor series expansion for the natural exponential function is that the Taylor series has a rather strict convergence domain, [17] making it difficult to converge in nonlinear systems. The Padé approximation, on the other hand, can solve this problem well, and for each delay, such a rational function structure is unique. [26] The construction principle of Padé approximation is: Given a natural number m and a positive integer n , the $[m, n]$ order Padé approximation of the function $f(x)$ is:

$$R(x) = \frac{\sum_{j=0}^m a_j x^j}{1 + \sum_{k=1}^n b_k x^k} = \frac{a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m}{1 + b_1 x + b_2 x^2 + \dots + b_n x^n} \quad (14)$$

And:

$$\begin{aligned} f(0) &= R(0) \\ f'(0) &= R'(0) \\ f''(0) &= R''(0) \\ &\vdots \\ f^{(m+n)}(0) &= R^{(m+n)}(0) \end{aligned} \quad (15)$$

However, the Padé approximation faces the issue of order selection. If the order is too low, the approximation error is large, making it difficult to estimate the delay accurately, thus poorly capturing the transition state. If the order is too high, although the error is smaller, it introduces high frequencies, which intensifies the oscillation of the observer and makes it difficult to converge. Therefore, the selection of the order for the Padé approximation is a key point. [26]

Another approximation method is to directly construct using the $\frac{as+b}{cs+d}$ structure. By minimizing the sum of the squares of the errors to find the most suitable parameters, an approximate rational function is constructed. This method is called the least squares method. It should be noted that although the least squares method has a higher approximation accuracy for delay functions than the Pade function, [18] it may not be able to capture the dynamic behavior of complex nonlinear systems. So, while the least squares method can better describe the transition state of the observed state, some approximations may not accurately estimate the amplitude. In this case, the Luenberger observer, which converges quickly, can be combined to adjust the gain. This can maximize the accuracy advantage of the least squares estimation, and also compensate for the overall amplitude loss caused by its inability to capture the complex dynamic behavior of nonlinear systems at times.

CHAPTER 3 The Observer Design

3.1 Linear Observer

3.1.1 Luenberger Observer

Due to the nonlinear characteristics of the control system, to use the Luenberger observer, the control system needs to be approximated as a linear system. Combining equations 3(b) and (5), the sigmoid functions $S_1(\cdot)$ and $S_2(\cdot)$ are simplified to proportional functions, and ignore the delay, resulting in:

$$\dot{x}_1(t) = \frac{1}{6}[-26x_1(t) - 10.7x_2(t) + 875] \quad (16a)$$

$$\dot{x}_2(t) = \frac{1}{14}[20x_1(t) - 13.3x_2(t)] \quad (16b)$$

And the output function is:

$$y = x_1(t) \quad (17)$$

Then the state matrix A, input matrix B, output matrix C, and output transfer matrix D of the above system are as follows:

$$A = \begin{bmatrix} \frac{-26}{6} & \frac{-10.7}{6} \\ \frac{20}{14} & \frac{-13.3}{14} \end{bmatrix} \quad B = \begin{bmatrix} \frac{875}{6} \\ 0 \end{bmatrix} \quad C = [1 \ 0] \quad D = 0$$

As mentioned in Section 2.3, it is necessary to find the L matrix by adjusting the eigenvalues of the A - LC matrix. A reasonable way to set the eigenvalues is to first find the eigenvalues of A, then enlarge the preset eigenvalues by 2 to 3 times based on the eigenvalues of A. You can set several groups of eigenvalues, compare their observation results, and finally lock in the optimal eigenvalues. This process can be more easily completed using Matlab's place function. It should be noted that in the place function, the transposes of the A and C matrices should be used, and the resulting L matrix also needs to be transposed again. And the B matrix of observer is $B = [B, L]$. Finally, $L = \begin{bmatrix} -2883.3 \\ 1752.4 \end{bmatrix}$.

And the result is in the following figure:

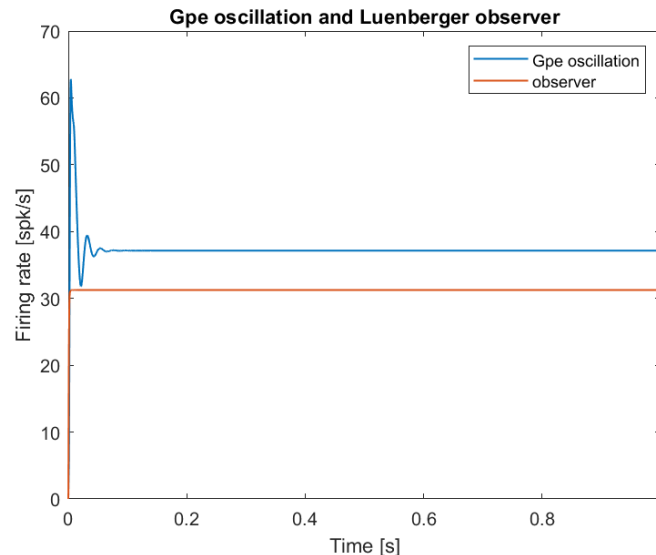


Figure4. Gpe oscillation and Luenberger observer

As shown in Figure4, although using the Luenberger observer can stabilize the final observation results, the stable values differ significantly from the actual state. Therefore, improvements to this observer are necessary. After numerous experiments, a critical input value was identified. The current reference value obtained by this controller is 35, with a total gain of 25, resulting in an input of 875. Through extensive experimentation, it was found that when the controller's reference value is less than or equal to 26, the observer's reference value should be increased by 9, and when the controller's reference value is greater than 26, the observer's reference value should be increased by 6. Specifically, set:

$$B = \begin{cases} \begin{bmatrix} \frac{25(\bar{x} + 6)}{6} \\ 0 \end{bmatrix}, & \bar{x} > 26 \\ \begin{bmatrix} \frac{25(\bar{x} + 9)}{6} \\ 0 \end{bmatrix}, & \bar{x} \leq 26 \end{cases} \quad (18)$$

The B matrix is the original B matrix in control system. Then obtain L using the above method of setting eigenvalues.

This approach yields stable values that are very close to the actual state. After adjusting, the result is as following figure:

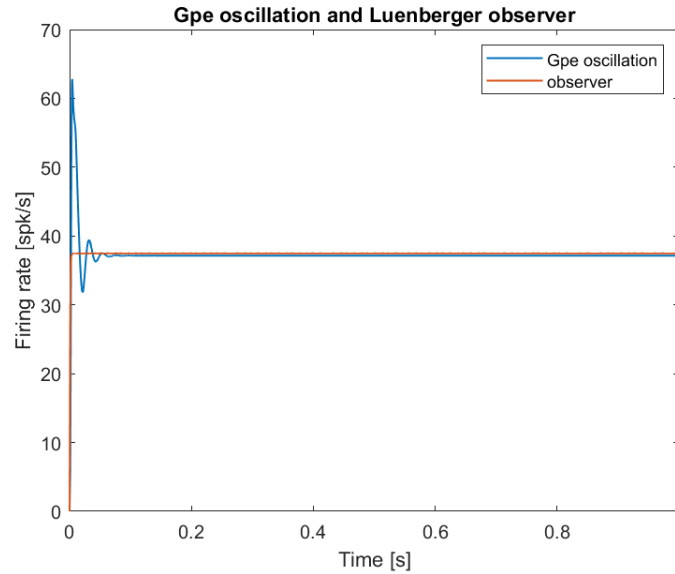


Figure5. Gpe oscillation and Luenberger observer

From Figure 5, it can be seen that such stable results are reasonable, but the transitional state is poor.

3.1.2 Luenberger observer with delay

From Table 1, we can obtain the specific delay data. If we consider these delay data, the linearization of the sigmoid function can be obtained as follows:

$$\dot{x}_1(t) = \frac{1}{6} [-26x_1(t) - 10.7x_2(t - 7) + 875] \quad (19a)$$

$$\dot{x}_2(t) = \frac{1}{14}[20x_1(t-1) - 13.3x_2(t-1)] \quad (19b)$$

And the output function is also same with the function (15). The delay of the above control system needs to be processed before it can be observed with the Luenberger observer. The first approximation used is the Padé approximation. Here are the specific steps to establish it:

1. Establish the delay terms $x_2(t-7)$, $x_1(t-1)$ and $x_2(t-1)$ as new states $x_3(t)$, $x_4(t)$, $x_5(t)$ respectively.
2. Use the Padé approximation to establish the equation for each state, which means that transfer the function from the time domain to frequency domain using Padé approximation and Laplace Transform.

Specifically: (use the first order of Padé approximation)

$$L\{x_2(t-7)\} = x_2(s) \times \frac{-s + \frac{2}{7}}{s + \frac{2}{7}} = x_3(s) \quad (20a)$$

$$L\{x_1(t-1)\} = x_1(s) \times \frac{-s + 2}{s + 2} = x_4(s) \quad (20b)$$

$$L\{x_2(t-1)\} = x_2(s) \times \frac{-s + 2}{s + 2} = x_5(s) \quad (20c)$$

And transfer the above functions in time domain again using the Laplace Transform:

$$\dot{x}_3(t) = \frac{-2}{7}x_3(t) - \dot{x}_2(t) + \frac{2}{7}x_2(t) \quad (21a)$$

$$\dot{x}_4(t) = -2x_4(t) - \dot{x}_1(t) + 2x_1(t) \quad (21b)$$

$$\dot{x}_5(t) = -2x_5(t) - \dot{x}_2(t) + 2x_2(t) \quad (21c)$$

3. Find the structure of the reconstructed control system again.

By combining equations (16a) and (16a), the final state equation can be derived:

$$\dot{x}_1(t) = -\frac{26}{6}x_1(t) - \frac{10.7}{6}x_3(t) + \frac{875}{6} \quad (22a)$$

$$\dot{x}_2(t) = -\frac{1}{14}x_2(t) + \frac{20}{14}x_4(t) - \frac{12.3}{14}x_5(t) \quad (22b)$$

$$\dot{x}_3(t) = (\frac{2}{7} + \frac{1}{14})x_2(t) - \frac{2}{7}x_3(t) - \frac{20}{14}x_4(t) + \frac{12.3}{14}x_5(t) \quad (22c)$$

$$\dot{x}_4(t) = \left(2 + \frac{26}{6}\right)x_1(t) + \frac{10.7}{6}x_3(t) - 2x_4(t) - \frac{875}{6} \quad (22d)$$

$$\dot{x}_5(t) = (2 + \frac{1}{14})x_2(t) - \frac{20}{14}x_4(t) + (\frac{12.3}{14} - 2)x_5(t) \quad (22e)$$

So based on the above functions, the state matrix A and input matrix B of the above system are as follows:

$$A = \begin{bmatrix} \frac{-26}{6} & 0 & \frac{-10.7}{6} & 0 & 0 \\ 0 & \frac{-1}{14} & 0 & \frac{20}{14} & \frac{-12.3}{14} \\ 0 & \frac{2}{7} + \frac{1}{14} & \frac{-2}{7} & \frac{-20}{14} & \frac{12.3}{14} \\ 2 + \frac{26}{6} & 0 & \frac{10.7}{6} & -2 & 0 \\ 0 & 2 + \frac{1}{14} & 0 & \frac{-20}{14} & -2 + \frac{12.3}{14} \end{bmatrix} \quad B = \begin{bmatrix} \frac{875}{6} \\ 0 \\ 0 \\ -\frac{875}{6} \\ 0 \end{bmatrix}$$

Refer to the method of deriving the L matrix in Section 3.1.1. The difference here is that, due to the introduction of delay estimation, new states are added, which expand the dimensions of the original A matrix, often resulting in paired complex eigenvalues. In this case, using the built-in Luenberger observer in Matlab's Simulink cannot preset the paired complex eigenvalues accordingly. If the paired complex eigenvalues are not preset, it will greatly reduce the final performance of the Luenberger observer, causing a mismatch with the actual control system speed. Therefore, when the A matrix has paired complex eigenvalues, it is best to manually set the paired eigenvalues and avoid directly using the built-in Luenberger observer in Matlab's Simulink, as this will not yield the best results.

See the figure below for the observation results:

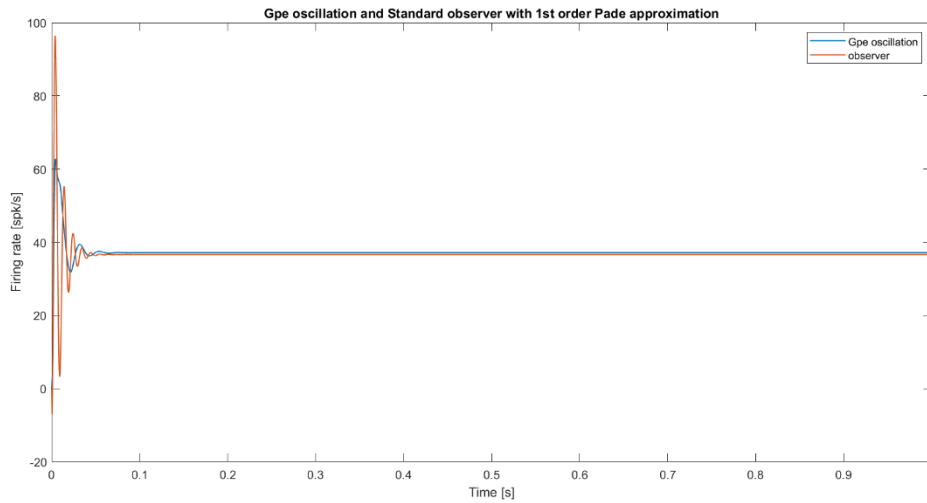


Figure6. Gpe oscillation and Luenberger observer with 1st order Padé approximation

From the above figure, it can be seen that the observer after considering the delay estimation captures the transition state much better than the one ignoring the delay.

However, the above is just a first-order Padé approximation, and its value has some differences from the actual delay. Therefore, considering the use of the second-order Padé approximation and using the same construction method as above, the results obtained are as follows:

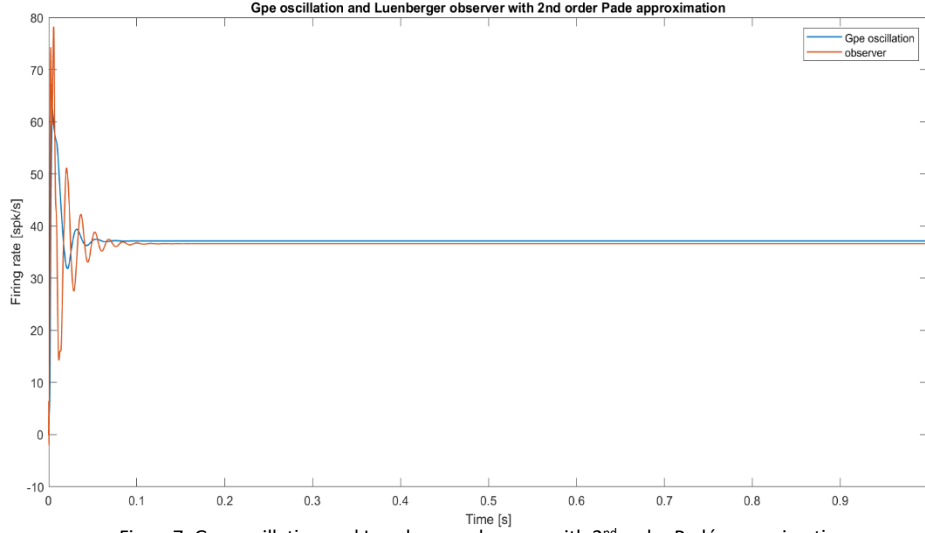


Figure7. GPe oscillation and Luenberger observer with 2nd order Padé approximation

From Figure 7, it can be seen that the second-order Padé approximation generates more oscillations in the transition state, but the frequency is already higher than the oscillation frequency of the original GPe state. [26] Therefore, the first-order Padé approximation is sufficient to capture the transition state.

As mentioned in section 2.4, the accuracy of the Padé approximation for delay approximation is far less than that of the least squares approximation. This is also why the Padé approximation can capture some transition state information, but it is not accurate. After increasing the order, although the accuracy increases, the oscillation frequency is prone to surge. [26] The most suitable rational form like 'a' is solved by using the 'lsqcurvefit' function in Matlab to estimate the delay.

The specific approximation data is as follows:

1st order:

$$e^{-s} \approx \frac{-2.4680s + 11.7045}{15.2607s + 10.9185} \quad (23a)$$

$$e^{-7s} \approx \frac{-0.3860s + 0.9285}{14.1137s + 0.8949} \quad (23b)$$

2nd order:

$$e^{-s} \approx \frac{0.4766s^2 - 5.5659s + 17.0410}{4.6892s^2 + 10.8017s + 17.0972} \quad (24a)$$

$$e^{-7s} \approx \frac{0.7490s^2 - 3.4752s + 2.6114}{71.8190s^2 + 12.9613s + 2.6087} \quad (24b)$$

The results obtained are as follows:

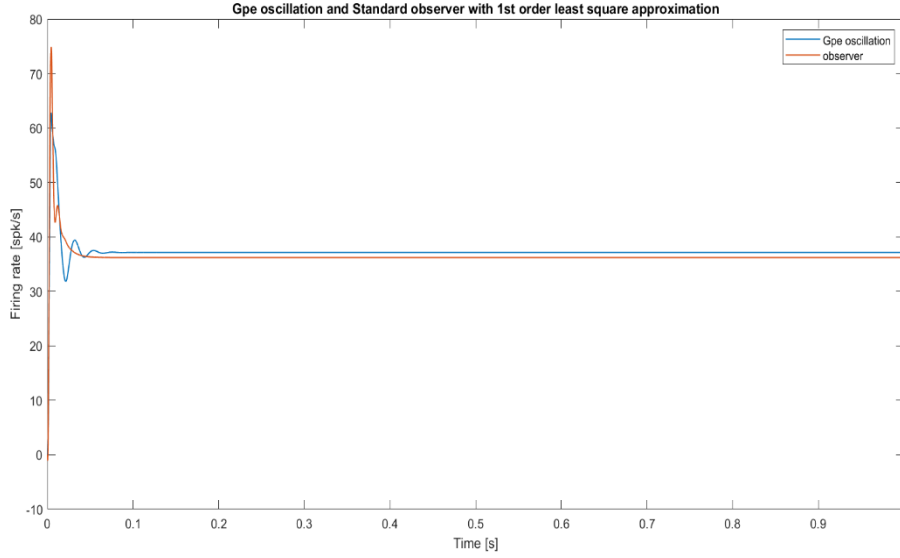


Figure8. GPe oscillation and Luenberger observer with 1st order least square approximation

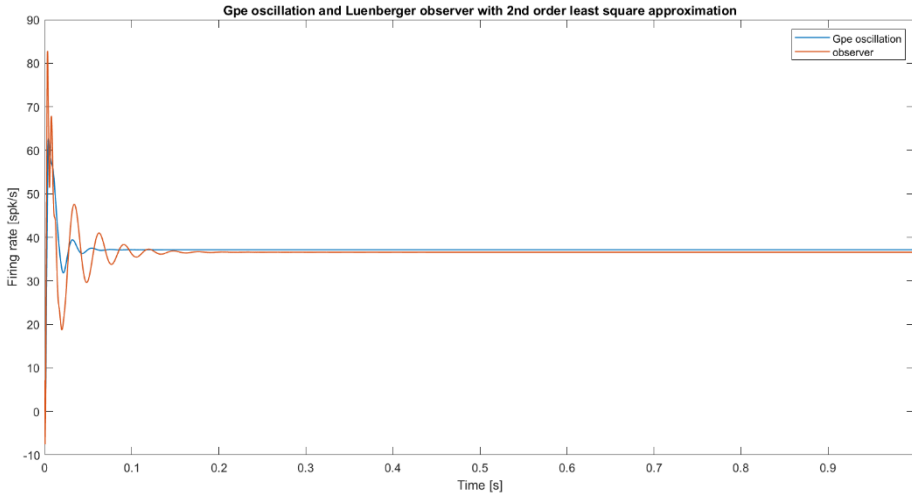


Figure9. GPe oscillation and Luenberger observer with 2nd order least square approximation

From Figure 8, it can be seen that the oscillation frequency of this method at the first order is lower than the oscillation frequency of GPe itself, so the second order needs to be considered. From Figure 9, it can be seen that the least squares approximation at the second order is significantly better than the Padé approximation at capturing the frequency of the transition state, but the oscillation amplitude is too large, which is also a problem that needs to be noted with the least squares method.

3.2 Nonlinear Observer

3.2.1 The Nonlinear Observer of Xingzhe Fan and Murat Arcak

According to formulas (11) and (12) in section 2.3, first, without considering delays, that is, ignoring delays, using the nonlinear observer construction method by Xingzhe Fan and Murat Arcak, [8] the final structure is:

$$\hat{\dot{x}}_1(t) = \frac{1}{6}[-y + S_1[-4.1667\hat{x}_1(t) - 10.7\hat{x}_2(t) - 20.8333]] \quad (25a)$$

$$\hat{x}_2(t) = \frac{1}{14}[-\hat{x}_1(t) - \hat{x}_2(t) + y + S_2[-4.5857\hat{x}_1(t) - 12.3\hat{x}_2(t) + 24.5857]] \quad (25b)$$

Obviously, $y = x_1(t)$.

It should be noted that solving the equations for the new observer mentioned above is not easy. It can be easily seen from the equations that they resemble another controller structure. The essence of Fan and Arcak's nonlinear observer lies in its clever design, which almost 'reproduces' the structure of the original controller. Solving for the L and K matrices in equation (11) by hand is evidently very difficult. Therefore, it is necessary to utilize Matlab's various functions for Linear Matrix Inequality (LMI) calculations. First, it is necessary to construct matrices that satisfy condition (12) to ensure that the observed state reaches stability.

In this design, set $L = \begin{bmatrix} a \\ b \end{bmatrix}$, $K = \begin{bmatrix} c \\ d \end{bmatrix}$, $P = \begin{bmatrix} e & f \\ f & h \end{bmatrix}$ and $\zeta = s$ using Matlab's `syms(·)` function to construct the system, the `assume(·)` function to assume these numbers are real, and the `eye(·)` function to construct the identity matrix, the specific matrices in (12) can be obtained as follows:

$$\begin{bmatrix} s + 2bf + 2e(a - \frac{1}{6}) & bh - \frac{f}{14} + f(a - \frac{1}{6}) & c + \frac{e}{6} - 25 & d + \frac{f}{14} + 20 \\ bh - \frac{f}{14} + f(a - \frac{1}{6}) & s - \frac{h}{7} & \frac{f}{6} - 10.7 & \frac{f}{14} - 12.3 \\ c + \frac{e}{6} - 25 & \frac{f}{6} - 10.7 & 0 & 0 \\ d + \frac{f}{14} + 20 & \frac{h}{14} - 12.3 & 0 & 0 \end{bmatrix}$$

The matrix above needs to satisfy the conditions in matrix (12), meaning its values need to be less than or equal to 0. Undoubtedly, solving for the values of L , K , and ζ above is challenging because, as shown in the matrix, L , K are matrices. Therefore, the only condition for the matrix values to be less than 0 actually requires deriving eight values: a, b, c, d, e, f, h , and s . This almost seems impossible! However, a possible solution is proposed here, which is a common mathematical approach from specific to general, which is the method of approximate inductive reasoning. Since the condition in (12) includes the condition of being equal to 0, we can first consider setting many elements in the matrix to 0, thus solving for some values. During this process, there is also the condition that matrix P must be greater than 0 and s must also be greater than 0, making it more likely to solve this matrix. Once all possible zeros are solved, by observing and fine-tuning, all values can eventually be solved. Of course, these values are not unique, this is one possible solution, and other values will ultimately achieve the stability when they also meets the above matrix conditions.

The result is as below figure:

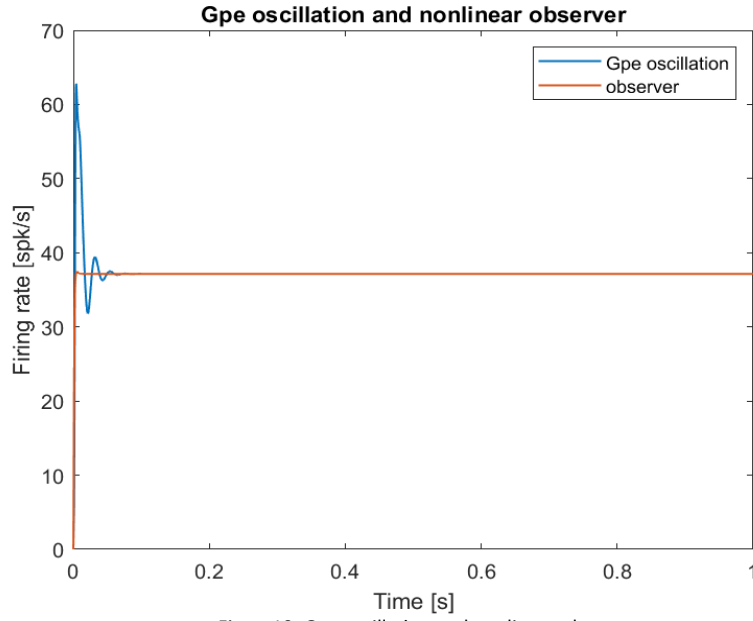


Figure10. Gpe oscillation and nonlinear observer

As shown in Figure 10, the nonlinear observer ignoring delays is very similar to the Luenberger observer ignoring delays, but it converges faster and the results are more accurate. However, it differs significantly from the Luenberger observer in that it can achieve a stable state close to the original state without setting an input threshold

3.2.2 Nonlinear observer with delay

If we consider the approximation of the Luenberger observer and delays, the results from section 2.1.2 show that a first-order Padé approximation is sufficient. Applying the first-order Padé approximation to the nonlinear observer, combining the function (12), we should let the following matrix's value is less or equal to 0:

$$\begin{bmatrix} s + 2c(a_1 - \frac{1}{6}) & a_2d & a_1c + a_3e & a_1c + f(a_4 + \frac{13}{6}) & a_1c + a_5g & b_1 + \frac{c}{6} - 25 & b_2 \\ a_2d & s - \frac{d}{7} & \frac{5e}{14} + a_2d & a_2d & \frac{29g}{14} + a_2d & 0 & \frac{d}{14} \\ a_1c + a_3e & \frac{5e}{14} + a_2d & s + 2e(a_3 - \frac{2}{7}) & a_3e + a_4f & a_3e + a_5g & b_1 - 10.7 & b_2 - \frac{e}{14} \\ a_1c + f(a_4 + \frac{13}{6}) & a_2d & a_3e + a_4f & s + 2f(a_4 - 2) & a_4f + a_5g & b_1 - \frac{f}{6} & b_2 + 20 \\ a_1c + a_5g & \frac{29g}{14} + a_2d & a_3e + a_5g & a_4f + a_5g & s + 2g(a_5 - 2) & b_1 & b_2 - \frac{g}{14} - 12.3 \\ b_1 + \frac{c}{6} - 25 & 0 & b_1 - 10.7 & b_1 - \frac{f}{6} & b_1 & 0 & 0 \\ b_2 & \frac{d}{14} & b_2 - \frac{e}{14} & b_2 + 20 & b_2 - \frac{g}{14} - 12.3 & 0 & 0 \end{bmatrix}$$

It should be noted that the matrices $L = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$, $K = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, $P = \begin{bmatrix} c & 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 & 0 \\ 0 & 0 & e & 0 & 0 \\ 0 & 0 & 0 & f & 0 \\ 0 & 0 & 0 & 0 & g \end{bmatrix}$ and $\zeta = s$

form the above matrix. Here, since P is a symmetric matrix greater than 0 and has dimensions of 5×5 , it will be a significant reason why the above matrix is difficult to solve. Therefore, P is specialized into a diagonal matrix because a diagonal matrix is always a symmetric matrix, and P also needs to satisfy the condition of being greater than 0.

It seems that the matrix above is very difficult to solve, but after observing carefully, many of its elements are repeated. Therefore, the method in 3.2.1 can still be used to solve some values first and then fine-tune the other values. However, it should be noted that the elements in the above matrix contain many elements that are pure variable multiplications. These elements cannot be solved in advance because to solve them in advance, one of the variables must be set to 0, which will cause many subsequent processing troubles.

And finally we obtain the following structure:

$$A = \begin{bmatrix} -0.1667 & 0 & 0 & 0 & 0 \\ 0 & -0.0714 & 0 & 0 & 0 \\ 0 & 0.3571 & -0.2857 & 0 & 0 \\ 2.1667 & 0 & 0 & -2 & 0 \\ 0 & 2.0714 & 0 & 0 & -2 \end{bmatrix} \quad C = [1 \quad 0 \quad 1 \quad 1 \quad 1]$$

$$H = \begin{bmatrix} 0.1667 & 0 \\ 0 & 0.0714 \\ 0 & -0.0714 \\ -0.1667 & 0 \\ 0 & -0.0714 \end{bmatrix} \quad L = \begin{bmatrix} 0.001 \\ 0.002 \\ 0.003 \\ 0.001 \\ 0.002 \end{bmatrix} \quad K = \begin{bmatrix} 0.2 \\ 0.18 \end{bmatrix}$$

And the result is as the following figure:

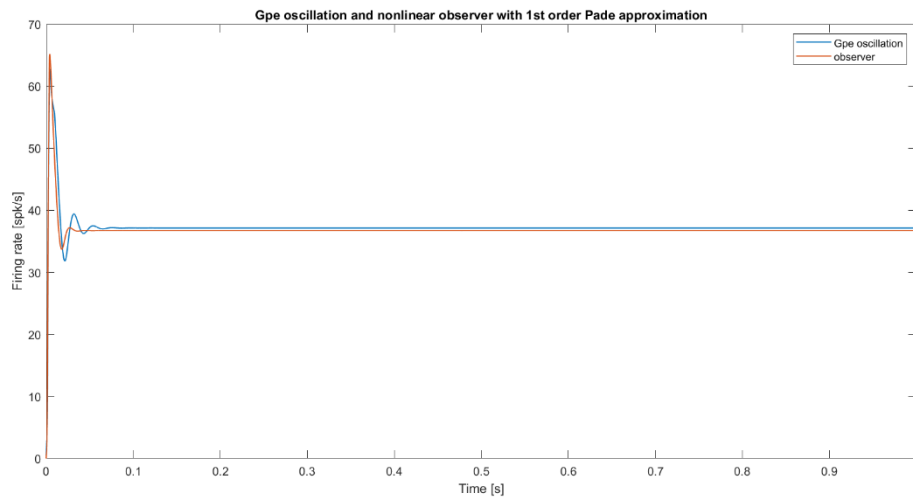


Figure11. Gpe oscillation and nonlinear observer with 1st Padé approximation

From the above figure, it can be seen that the advantage of the nonlinear observer appears after adding the Padé approximation of the delay. Compared to Figure 6, this advantage is even more obvious. When the Padé approximation is applied to the Luenberger observer, the inaccuracy of the delay approximation is further amplified due to the simplification of the system to a linear system. [25] However, in a nonlinear system, the impact of this inaccuracy is significantly reduced. Therefore, in Figure 20, it can be observed that the combination of the Padé approximation and the nonlinear observer greatly improves the accuracy of capturing the transitional state of GPe.

When using the first-order least squares approximation for delays in combination with the nonlinear observer, the appropriate structure is as follows:(The method of solving is same as the above discussion)

$$A = \begin{bmatrix} -4.3333 & 0 & -1.7833 & 0 & 0 \\ 0 & -0.0714 & 0 & 1.4286 & -0.8786 \\ 0 & 0.0677 & -0.0634 & 0 & 0 \\ 0.7942 & 0 & 0 & -0.7159 & 0 \\ 0 & 0.7788 & 0 & 0 & -0.7159 \end{bmatrix} \quad C = [1 \quad 0 \quad 1 \quad 1 \quad 1]$$

$$H = \begin{bmatrix} 0.1667 & 0 \\ 0 & 0.0714 \\ 0 & -0.002 \\ -0.027 & 0 \\ 0 & -0.0116 \end{bmatrix} \quad L = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \\ 0.1 \\ 0.2 \end{bmatrix} \quad K = \begin{bmatrix} 0.2 \\ 0.18 \end{bmatrix}$$

And the result is shown in the Figure 21:

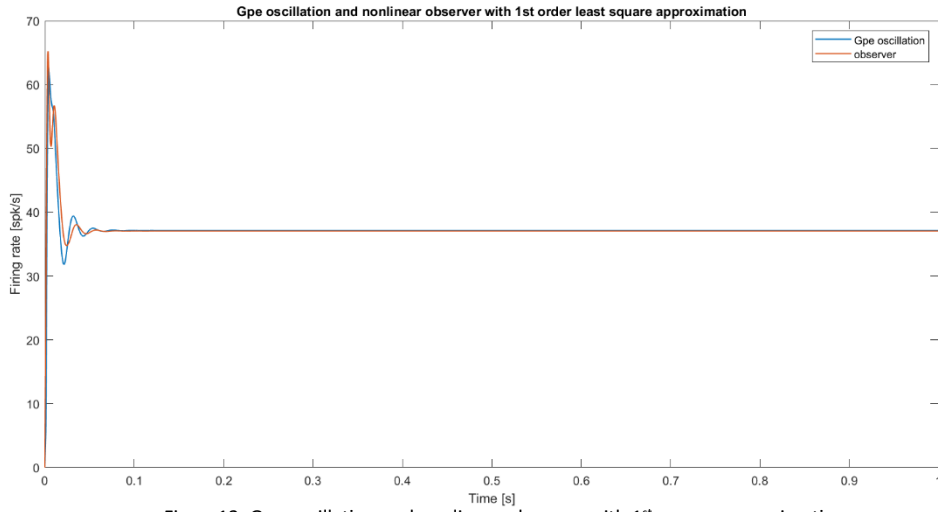


Figure12. GPe oscillation and nonlinear observer with 1st square approximation

Comparing Figure 12 with Figure 11, it can be seen that the results of the least squares approximation are superior in tracking the transitional state in the nonlinear observer. This is because the least squares method estimates delays more accurately than the Padé approximation, [24] and this advantage is more evident when applied to nonlinear systems. From Figure 12, it can be observed that the oscillations of GPe in the transitional state are captured at every peak, both upward and downward.

3.3 Advanced Nonlinear Observer

From the above discussion, it can be seen that the inaccuracy of delay estimation and the resulting overestimation or underestimation of frequency can cause significant loss in the ability to capture the GP transition state. Therefore, if a nonlinear observer that does not require delay estimation can be designed, this part of the loss can be minimized.

The model of the control system that this project complies with is:[8]

$$\dot{x} = Ax + G\theta(Hx(t - \alpha)) + \mu(y, u) \quad (26a)$$

$$y = Cx \quad (26b)$$

where $x \in R^n$ is the state, $y \in R^r$ is the measured output, $u \in R^m$ is the control input, and the multivariable nonlinearity $\theta(\cdot)$ satisfy a multivariable analog of the monotonicity property:[8]

$$\frac{\partial \theta}{\partial v} + \frac{\partial \theta^T}{\partial v} \geq 0, \forall v \in \mathbb{R}^p \quad (27)$$

With this assumption, the observer has the form:

$$\dot{\hat{x}} = A\hat{x} + L[C\hat{x} - y] + G\theta[H\hat{x}(t - \alpha) + K[C\hat{x}(t - \alpha) - y(t - \alpha)]] + \mu(y, u) \quad (28)$$

The task is to design the observer matrices $K \in \mathbb{R}^{p \times r}$ and $L \in \mathbb{R}^{n \times r}$ to make the observer error $e = x - \hat{x}$ approach zero. From (23a) and (24) the dynamics of the observer error are governed by: $\dot{e} = (A + LC)e + G[\theta(m(t - \alpha)) - \theta(w(t - \alpha))]$ (29)

$$\text{where: } m(t) = Hx(t), w(t) = H\hat{x}(t) - KCe(t) \quad (30)$$

According to the Lyapunov function, [19]

$$V = x^T P x + \int_{t-\alpha}^{\alpha} x(\varepsilon)^T S x(\varepsilon) d\varepsilon \quad (31)$$

And the Leibniz Newborn Formula: [20]

$$Z(t - \alpha) = z(t) - \int_{t-\alpha}^t \dot{Z}(t + \beta) d\beta \quad (32)$$

So, the differentiated Lyapunov function [21]

$$\dot{V} = 2x^T P \dot{x} + x^T S x - x(t - \alpha)^T S x(t - \alpha) \quad (33)$$

in the system of the project:

$$\dot{V}(e) = 2e^T P \dot{e} + e^T S e - e(t - \alpha)^T S e(t - \alpha) \quad (34)$$

Combine the function (26), and reconstruct a part of it to the following function:

$$\theta(m(t - \alpha)) - \theta(w(t - \alpha)) = \phi(m, w) = \theta(m) - \theta(w) \quad (35)$$

The function (31) can be written as the function below:

$$\dot{V}(e) = 2e^T P[(A + LC)e + G\phi(m, w)] + e^T S e - e(t - \alpha)^T S e(t - \alpha) \quad (36)$$

So:

$$\dot{V}(e) \leq \begin{bmatrix} e \\ e(t - \alpha) \\ \phi(m, w) \end{bmatrix}^T \begin{bmatrix} (A + LC)^T P + P(A + LC) + S & 0 & PG \\ 0 & -S & (H + KC)^T W \\ 0 & 0 & -2W \end{bmatrix} \begin{bmatrix} e \\ e(t - \alpha) \\ \phi(m, w) \end{bmatrix} \quad (37)$$

where $P = P^T > 0$, $S = S^T > 0$.

So, the condition for asymptotically stable is:

$$\begin{bmatrix} (A + LC)^T P + P(A + LC) + S & 0 & PG \\ 0 & -S & (H + KC)^T W \\ 0 & 0 & -2W \end{bmatrix} < 0 \quad (38)$$

Due to the fewer variables, solving the above matrix is not more difficult than solving the matrix in 3.2.2. Moreover, a simpler approach is to set the elements in L and K to be the same in each column, which greatly reduces the number of variables.

With the above functions, finally the matrices solved for the new nonlinear observer is:

$$L = \begin{bmatrix} 0.001 \\ 0.001 \end{bmatrix} \quad K = \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix}$$

And the function of the advanced nonlinear observer is: $\dot{\hat{x}}_1(t) = \left(-\frac{1}{6} + 0.001\right) \hat{x}_1 +$

$$\frac{1}{6} S_1 [-25\hat{x}_1 - 10.7\hat{x}_2(t-7) + 0.2\hat{x}_1(t-1) - 0.2y(t-1)] - 0.001y \quad 39(a)$$

$$\dot{\hat{x}}_2(t) = 0.001\hat{x}_1 - \frac{1}{14} \hat{x}_2 +$$

$$\frac{1}{14} S_2 [20.2\hat{x}_1(t-1) - 12.3\hat{x}_2(t-1) - 0.2y(t-1)] - 0.001y \quad 39(b)$$

And the expression of y is same as the function 7(b), and the $C = [1, 0]$.

Based on the above functions, the result is in the following figure:

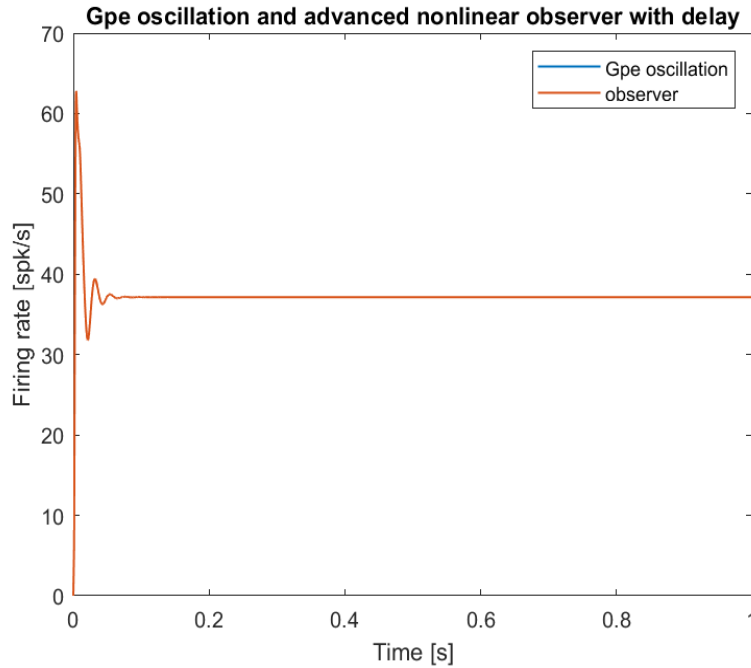


Figure13. Gpe oscillation and advanced nonlinear observer with delay

The Figure13 shows that the observed GPe state aligns perfectly with its actual state on the graph. However, when zoomed in using MATLAB's plotting tools, slight differences between the curves can be seen, which is consistent with the conclusions derived earlier. Clearly, this observer performs better than all the previous ones.

All discussions starting from Section 3.1 are based on the model of proportional feedback self-regulation. However, in Section 2.2, a more practical parameter self-tuning model is also discussed, where the value of proportional feedback is automatically adjusted according to the actual state of STN. [5] In this model, it is clear that the observer cannot be a Luenberger observer. Nonlinear observers are also difficult to implement due to the nonlinearity of the parameter self-tuning equation in Section 6(b). Therefore, for the parameter self-tuning model, the most suitable observer can only be an advanced nonlinear observer. We can assume that: the form of $\mu(y, u)$ in 23(a) is like:

$$\mu(y, u) = f(x_1(t))x_1(t) \quad (40)$$

Therefore, the self-tuning parameter $f(x_1(t))$ is only related to the input $x_1(t)$. When the observed value of $x_1(t)$ is close to accurate, the self-tuning parameter $f(x_1(t))$ will inevitably be accurate without delay. Thus, it can be equivalent to a constant, a "constant" that changes

accurately. Therefore, the advanced nonlinear model derived earlier can still be used, and the final result is as follows:

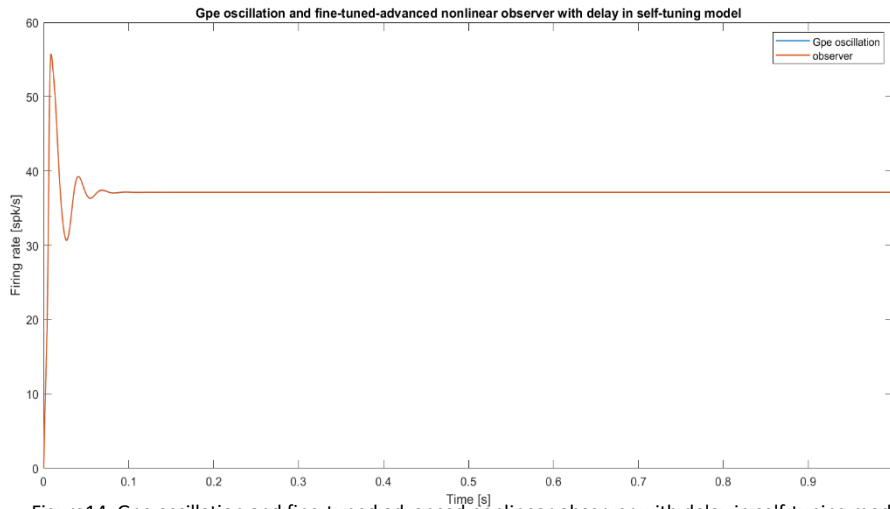


Figure14. GPe oscillation and fine-tuned advanced nonlinear observer with delay in self-tuning model

From Figure 14, it can be seen that the advanced nonlinear observer still performs well in the parameter adaptive model.

CHAPTER 4 Performance analysis of the Observers

4.1 Overview of Observer Evaluation Angles

It is important to emphasise here that the observer does not need to resort to any prior foreknowledge of the data, but rather observes the unknown ground state in real time, which is its unique advantage. However, this also makes it difficult to observe accurately. Based on the actual GPe output state of this project, the analysis of the observer's performance can be divided into two major aspects. One is the observer's performance during the transition state of GPe, and the other is its performance in the steady state. [22] From the perspective of the observer itself, there are two angles of analysis: accuracy and robustness. Combining the transition state of GPe and the perspective of the observer itself, four dimensions of observer performance analysis can be derived: accuracy in the steady state, robustness in the steady state, accuracy in the transition state, and robustness in the transition state. [24] However, because the accuracy of the values during the transition state is not a major focus in practice, and there is more attention to its dynamic state, the robustness of the transition state is temporarily not considered. The accuracy in the steady state is measured by the ratio of the observer's steady-state value to the actual steady-state value, which is the closer to 1, the better the performance. The robustness of the steady state is measured by the RMSE (Root Mean Square Error) of the steady state after adding Gaussian white noise; the lower the RMSE, [24] the better the robustness. The accuracy in the transition state mainly depends on whether the observer can

capture the actual actions during the transition state. [22] Therefore, it is measured from two angles. The first is the extent of tracking the macro movements, which is measured by the correlation coefficient between the transition states; the closer the coefficient is to 1, the better the movement tracking. The second is the capture of key movements, such as peaks, which is determined by the ratio of the number of peaks actually captured by the observer to the number of actual peaks in the GPe transition state. The closer the ratio is to 1, the better the capturing effect.

4.2 Observation Figure after Adding Noise

Since there is no literature that evaluates the performance of controllers for the oscillations between STN and GPe by adding Gaussian white noise, [28] and considering that the noise involving human brain neurons is not significant, Gaussian white noise with a variance of 0.1 is added to the input. The state diagrams of each observer are as follows: (Some observers cannot maintain stability, and their state figures are not shown below).

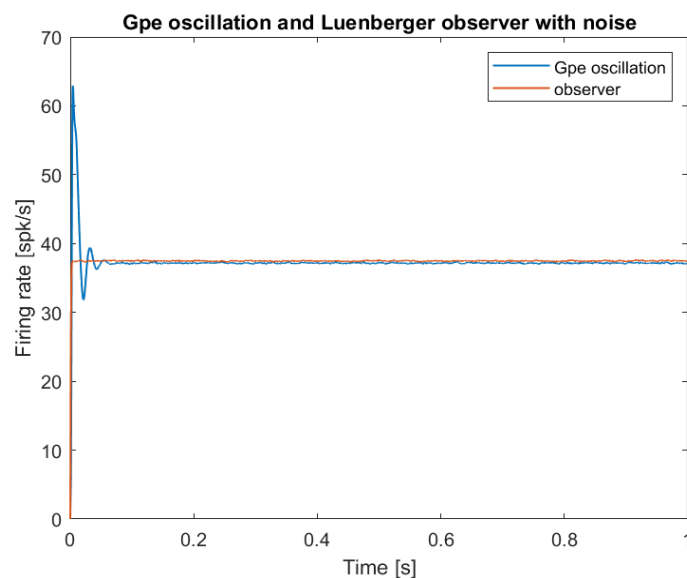


Figure15. Gpe oscillation and Luenberger observer with noise

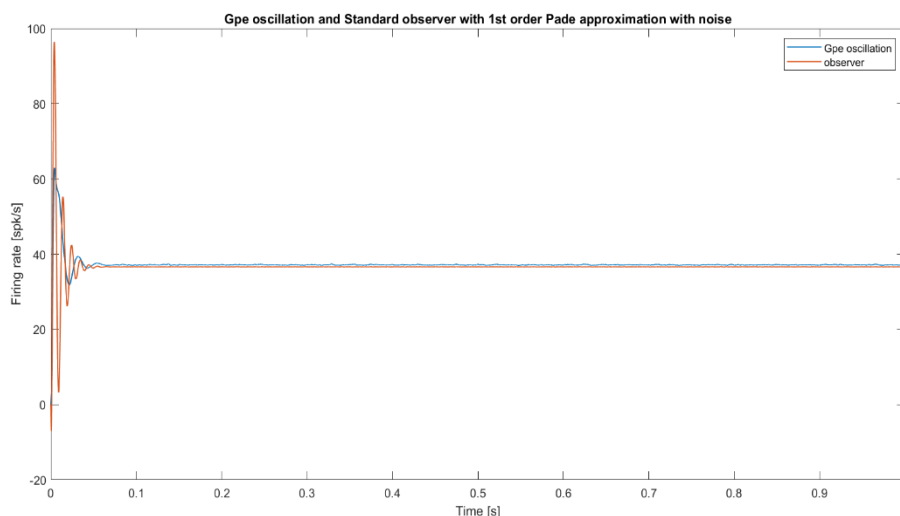


Figure16. Gpe oscillation and Standard observer with 1st order Padé approximation with noise

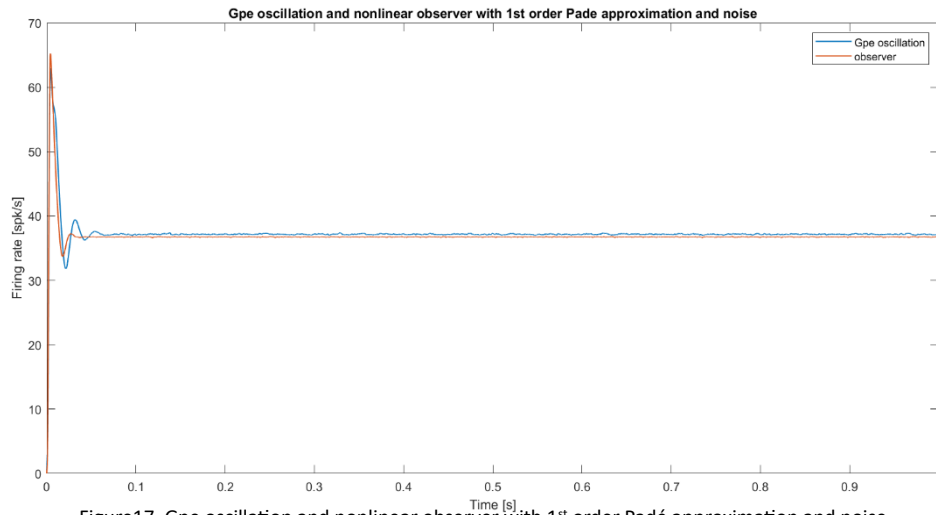


Figure17. Gpe oscillation and nonlinear observer with 1st order Padé approximation and noise

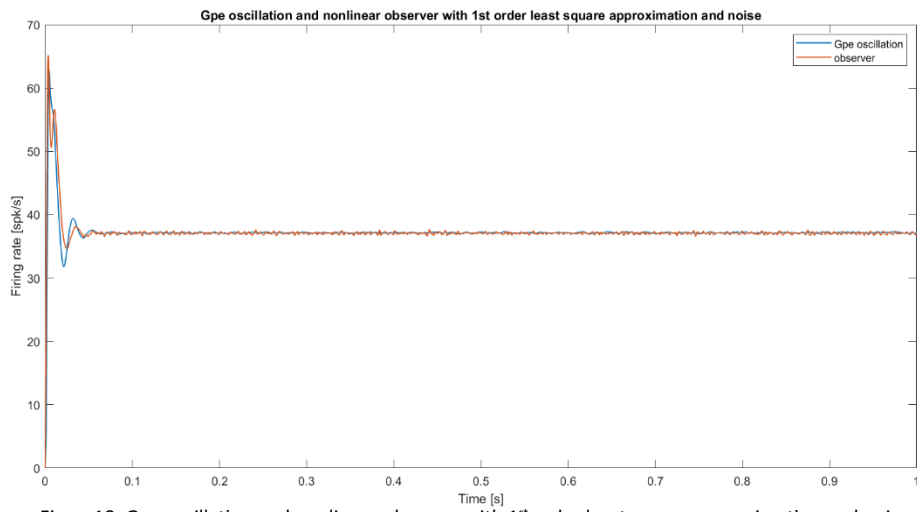


Figure18. Gpe oscillation and nonlinear observer with 1st order least square approximation and noise

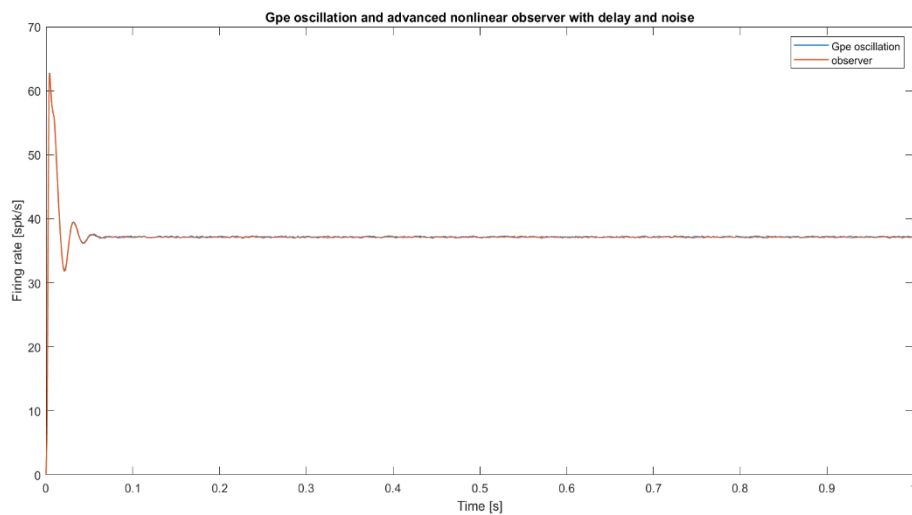


Figure19. Gpe oscillation and advanced nonlinear observer with delay and noise

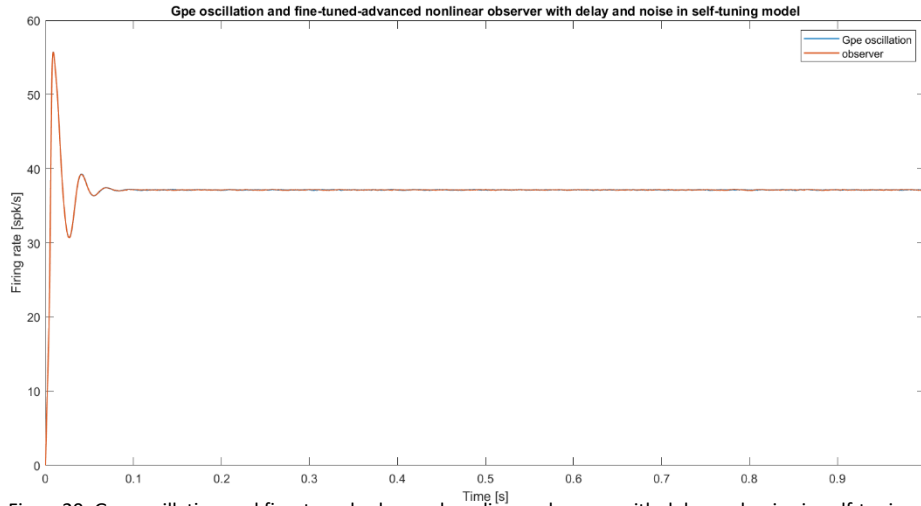


Figure20. Gpe oscillation and fine-tuned-advanced nonlinear observer with delay and noise in self-tuning model

4.3 Performance Analysis Table

TABLE 2 PERFORMANCE ANALYSIS TABLE

	Accuracy	Correlation Coefficient	Peak-to-Peak ratio	RMSE
Luenberger observer	0.9911	0.5098	0	0.3252
Luenberger observer with 1st Padé approximation	0.9902	0.4432	0.5714	0.3547
Luenberger observer with 2nd Padé approximation	0.9902	0.3776	0.2857	1.4884
Luenberger observer with 1st LSM approximation	0.9985	0.8862	0.1429	0.2292
Luenberger observer with 2nd LSM approximation	0.9914	0.7606	0.4286	
Nonlinear observer	1	0.5541	0.1429	
Nonlinear observer with 1st Padé approximation	0.9888	0.9560	0.4286	0.6495
Nonlinear observer with 1st LSM approximation	0.9975	0.8609	1	0.3134
Advanced nonlinear observer	1	1	1	0.05
Fine-tuned-advanced nonlinear observer	1	1	1	0.0242

The actual meaning of each indicator in the above table will be detailed in Chapter 5.

CHAPTER 5 Discussions and Conclusions

5.1 Accuracy

From Table 2, it can be seen that the accuracy of the stable state of all observers is above 0.9, proving that their performance in this aspect is quite good. Among them, the most outstanding performance is nonlinear observer, advanced observer and fine-tuned-advanced nonlinear observer, with an accuracy of 1, equivalent to completely accurate. This is a very good result.

For the nonlinear observer that does not consider delay, its stable state value is the same as the actual state, proving that the delay has little impact on the stable state value. Many nonlinear observers that perform delay estimation do not achieve an accuracy of 1, indicating that errors in delay estimation can affect accuracy. This provides another idea for DBS stimulation, as the convergence speed of the nonlinear observer is too fast, and it has already converged in the transition state of actual stimulation. This indicates that the final stable state value can be determined through the nonlinear observer, which is very useful for adjusting DBS stimulation. However, the structure of the nonlinear observer is obviously more complex than the Luenberger observer. [24] Considering the cost, the Luenberger observer can also be used to replace this function. Although the accuracy is slightly lower, 0.9911 is still very high, and the cost will be much lower. Observers with an accuracy of 1 also include the advanced nonlinear observer and the fine-tuned advanced nonlinear observer. These two have very high accuracy because they do not need to estimate the delay, and their structure is only slightly more complex than the nonlinear observer. The performance of these two observers will stand out in the subsequent performance analysis.

5.2 Correlation Coefficient

In this section of the evaluation, the Advanced Nonlinear Observer and the Fine-tuned Advanced Nonlinear Observer are the best performers, both having a transition state correlation coefficient of 1. This indicates that these observers capture the transition states most effectively, whether in a system with feedback or a self-tuning parameter system. [23] This aligns with previous inferences, as the advanced nonlinear observer not only has a nonlinear structure but also does not estimate delays, thus avoiding the amplification of errors caused by delay estimation in nonlinear systems. This makes it particularly effective in capturing unstable states. The Nonlinear Observer with 1st Padé Approximation has the second highest correlation coefficient, indicating that while approximating delays introduces errors, nonlinear observers are much better at tracking oscillations than linear ones. [24] The Luenberger Observer with 1st LSM Approximation has the third highest correlation coefficient, [25] showing that even in linear observers, the accuracy of LSM approximation in estimating delays can compensate for some of the limitations of linear observers.

5.3 Peak-to-Peak Ratio

In terms of peak-to-peak ratio performance, the best performers are the Nonlinear Observer with 1st LSM Approximation, Advanced Nonlinear Observer, and Fine-tuned Advanced Nonlinear Observer. The peak-to-peak ratio for the Advanced Nonlinear Observer and Fine-tuned Advanced Nonlinear Observer is 1, for the same reasons analysed in section 5.2. The Nonlinear Observer with 1st LSM Approximation also has a ratio of 1, indicating that although delay estimation errors can affect the observer's tracking of transition states, it is still possible to achieve the same effect in capturing critical information as the Advanced Nonlinear Observer. However, the Nonlinear Observer with 1st LSM Approximation has a lower correlation coefficient compared to the Nonlinear Observer with 1st Padé Approximation. [26] This is a potential drawback of LSM estimation: while it provides more accurate delay estimation, it can cause amplitude inaccuracies in nonlinear systems. Therefore, although the Nonlinear Observer with 1st LSM Approximation is very good at capturing critical information of transition states,

its correlation coefficient with the original state is not necessarily better than that of the Nonlinear Observer with 1st Approximation due to amplitude variations.

5.4 RMSE

RMSE measures the robustness of the observer against noise interference. [27] After adding Gaussian white noise, the steady-state value of the original system cannot remain completely stable and fluctuates with the variance of the noise. [28] This part requires higher performance considerations for the observer. Among all the tested observers, the one with the smallest RMSE is the Fine-tuned-advanced nonlinear observer. Theoretically, as long as the RMSE is less than $\sqrt{0.1} \approx 0.3162$, it indicates that the observer's robustness is qualified. However, the Fine-tuned-advanced nonlinear observer has an RMSE far less than 0.3162, which is 0.0242, indicating excellent noise resistance and robustness. The second smallest is the Advanced nonlinear observer, with an RMSE of 0.05. Although slightly larger than the Fine-tuned-advanced nonlinear observer, it still has good robustness. The reason for its slightly lower robustness is that its controller performance is weaker than that of the Fine-tuned-advanced nonlinear observer. The Fine-tuned-advanced nonlinear observer uses an adaptive parameter controller, while the Advanced nonlinear observer uses a proportional feedback controller, which also explains why the adaptive parameter model observer has better noise resistance. The adaptive parameter model not only extends the controller's adaptability in complex environments but also benefits the observer's robustness against noise. The Luenberger observer with 2nd LSM approximation cannot stabilize after adding noise, indicating that improving the accuracy of delay approximation can lead to frequency increase issues. These frequency disturbances can be linearly amplified in linear systems, potentially leading to non-convergence after adding noise. This is a critical issue to be addressed in delay approximation.

5.5 The best observer's performance with changing C_{ij}

As mentioned in Section 2.1, the constants $c_{ij} \geq 0$ denote the strength of synaptic connections from population j to population i . And all c_{ij} can be expressed in the form $(c_{ij} = a_{ij} + b_{ij}k)$, where a_{ij} and b_{ij} represents an inherent constant, which is the value for a normal person, and k is a variable ranging from 0 to 1. Among them, a_{12} , a_{21} and a_{22} are 1.1, 19, and 6.6 respectively, while b_{12} , b_{21} and b_{22} are 9.6, 1, and 5.7 respectively. The variation of k from 0 to 1 represents the severity of Parkinson's disease, with 0 indicating normal and 1 indicating fully diseased. This report primarily focuses on the fully diseased condition, evaluating the performance of the best observer, namely the fine-tuned advanced nonlinear observer, across different levels of Parkinson's disease severity. The experimental results are as follows:

1. The accuracies of stable state without adding noise are always 1.
2. The Correlation Coefficients of transitional state without adding noise are always 1.
3. The Peak-to-Peak ratios of transitional state without adding noise are always 1.

After adding AWGN:

1. The RMSE of the different c_{ij} is shown in the following figures:

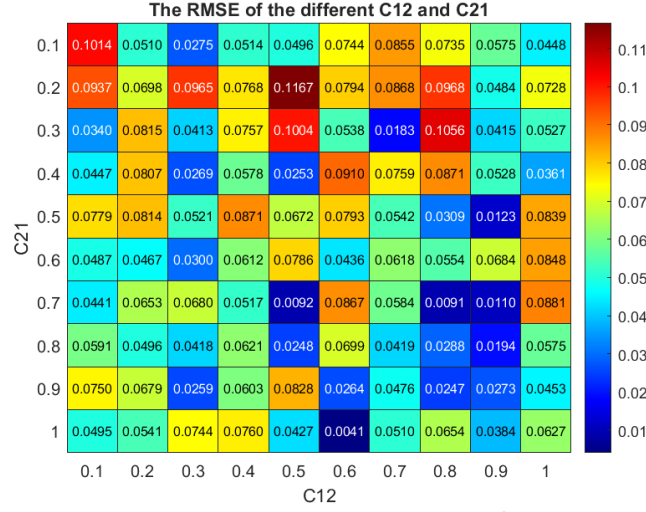


Figure21. The RMSE of the different c_{12} and c_{21}

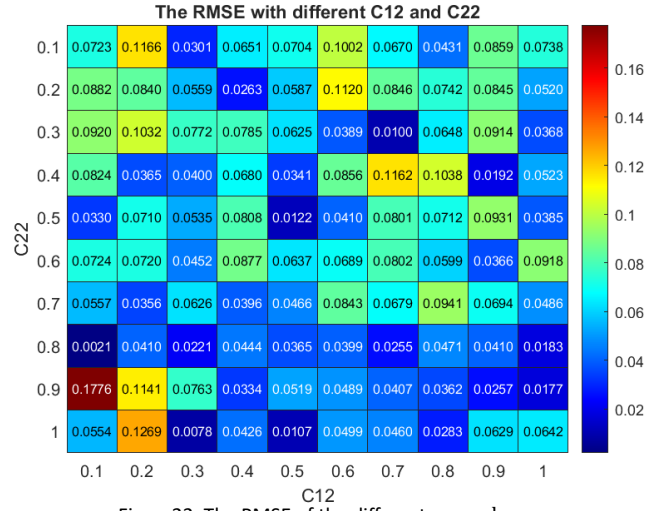


Figure22. The RMSE of the different c_{12} and c_{22}

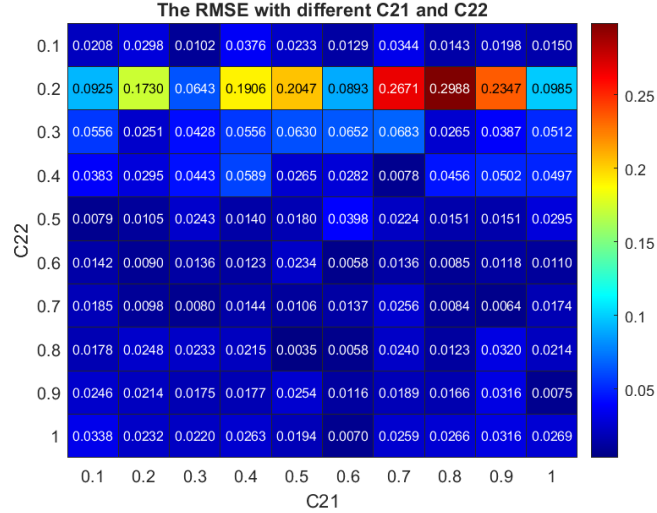


Figure23. The RMSE of the different c_{21} and c_{22}

And the mean RMSE of the different c_{12} and c_{21} is 0.0579, c_{12} and c_{22} is 0.0603 and c_{12} and c_{22} is 0.0388, which are all also far less than $\sqrt{0.1} \approx 0.3162$, which is the standard deviation of AWGN.

2. The Peak-to-Peak ratios of transitional state are always 1.

3. The Correlation Coefficients of transitional state with different c_{ij} is shown in the following figures:

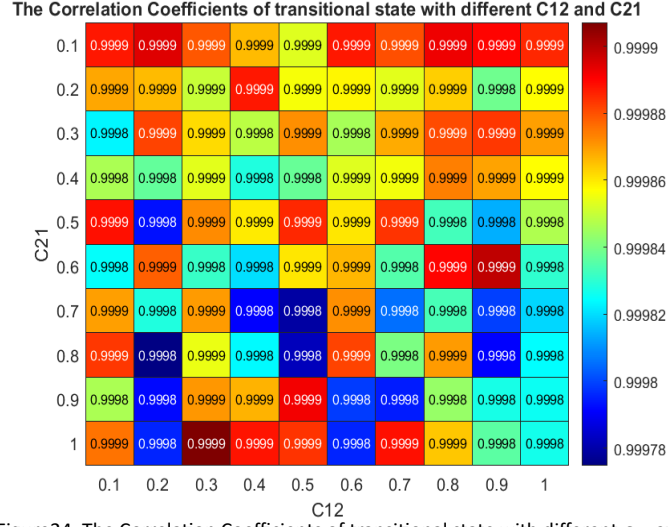


Figure24. The Correlation Coefficients of transitional state with different c_{12} and c_{21}

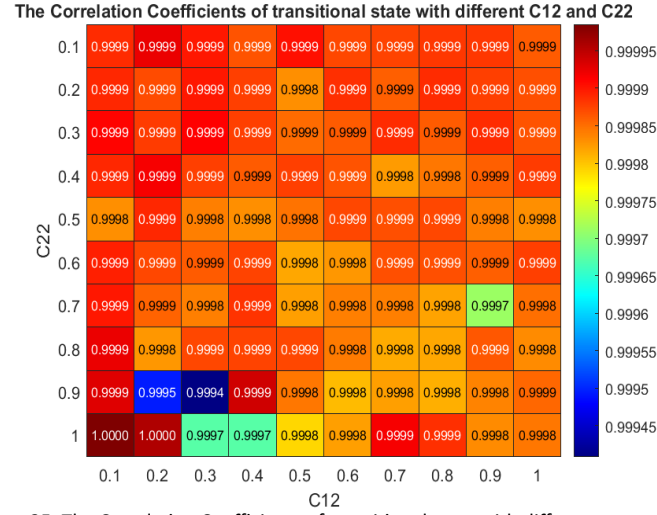


Figure25. The Correlation Coefficients of transitional state with different c_{12} and c_{22}

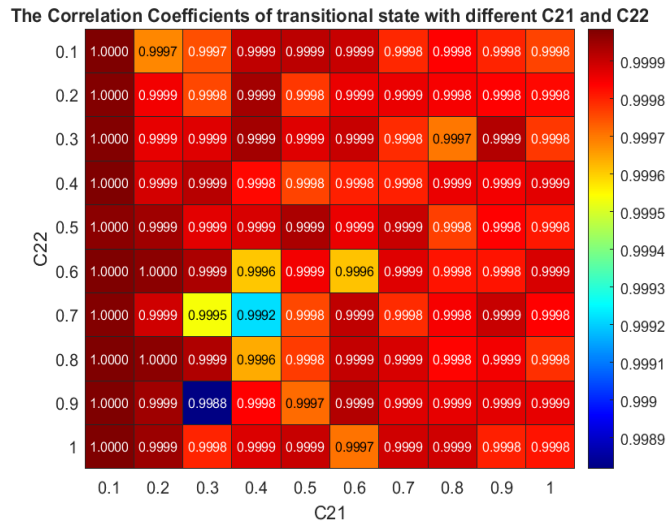


Figure26. The Correlation Coefficients of transitional state with different c_{21} and c_{22}

And the mean Correlation Coefficients of transitional state with different c_{ij} is 0.999.

The above assessment demonstrates that the observer is very effective in observing the state of the GPe in patients with varying degrees of Parkinson's disease, proving the transferability of the observer.

5.6 Conclusions

Overall, this report designs different observers to monitor the state of the GPe based on the Firing rate model. Firstly, the model is simplified to a linear model, and the traditional Luenberger observer is used, ignoring delays, resulting in a stable outcome as shown in Figure 3, but with a significant deviation from the actual stable value of the GPe. After extensive experiments, the relationship between the required input for the observer and the actual input was obtained, solving the error problem in the stable state of the system caused by the traditional Luenberger observer ignoring delays.

Next, the multi-state nonlinear observer proposed by Fan and Arcak was used, achieving faster convergence than the Luenberger observer and without the need to consider changing the actual input. Furthermore, this report considers the impact of delays on the observer. Delays are handled in two ways: firstly, by estimating the delay. This part is divided into two methods, namely Padé approximation and least squares method. Padé approximation can accurately capture the amplitude of various states, while the least squares method is superior in capturing the trend of transitional states. These two delay estimation methods were incorporated into the Luenberger observer and the nonlinear observer, resulting in various observation results detailed in Chapter 5.

The second method of handling delays is to retain them directly and construct a new type of nonlinear observer concerning the delay, referred to as the advanced nonlinear observer in this report, detailed in Section 3.3. This observer significantly outperforms the previously mentioned observers in terms of accuracy, ability to capture the transitional state of the actual GPe, and robustness. Finally, this report upgrades the advanced proportional feedback control model to an adaptive feedback control model, where the feedback parameters can automatically change with the STN state of the system. By applying the best advanced nonlinear observer, a fine-tuned advanced nonlinear observer was obtained, resulting in the best-performing observer. Not only does it achieve 100% accuracy in the stable state, but it also captures the transitional state very well, tracking all peak values of the actual state. The trend of the observed transitional state is completely positively correlated with the actual transitional state trend. Most importantly, its robustness is excellent, with an RMSE of only 0.0242 when the STN state input to the observer is disturbed by Gaussian white noise, demonstrating its excellent noise resistance.

In summary, the Fine-tuned-advanced nonlinear observer has the best performance evaluation in every aspect among all observers, making it the preferred observer for GPe observers. In terms of complexity, the Fine-tuned-advanced nonlinear observer is simpler than all nonlinear and linear observers that require delay estimation, as it eliminates the need to establish a new state for delay estimation. The Fine-tuned-advanced nonlinear observer is slightly more complex than the advanced nonlinear observer, with the addition of adaptive adjustment parameters. Therefore, in cases where the patient's condition is relatively stable, the advanced nonlinear observer is the best choice for good performance and low cost. However, in situations where the patient's condition or environmental changes cause system instability, the Fine-tuned-advanced nonlinear observer is the best choice. As for other observation methods, when

ignoring delays, both nonlinear observers and Luenberger observers can quickly converge to the actual stable state or an approximate stable state, providing insights for DBS treatment. These observers have the function of predicting stable state values, which can provide information for adjusting DBS stimulation. Among them, nonlinear observers can achieve zero-error prediction but are slightly more complex than Luenberger observers. Luenberger observers can achieve an accuracy of stable state of over 99%, making them a cost-effective choice when cost considerations are paramount and the demand focus on the stable state. However, in this project, although it is ultimately not necessary to estimate the delay, the value of the delay is assumed to be known in advance. Future research might consider how to improve the observer to adapt to unknown delays.

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